MARCUS DIBATTISTA

Engineering Leader & Cyber-Physical Systems Specialist

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Atlanta Metropolitan Area

Dear Hiring Manager,

As a strategic builder and engineering leader specializing in **Cyber-Physical Systems (CPS)** and **AI-driven data infrastructure**, I am writing to express my strong interest in joining your team with a goal of engineering research, innovation, and process improvement management. My professional framework centers on a single, critical objective: bridging complex physical sensing layers with advanced ML pipelines to architect reliable, scalable, modular, high-performance hardware & software solutions while empowering and personally developing the teams that build them.

Currently with a Ph.D. in Mechanical Engineering at the University of Georgia, my dissertation work focused on a modular co-design framework for advanced monitoring of cyber-physical assembly testbeds and time-series video entropy extraction algorithms. I drive deep technical execution across university research and advanced manufacturing initiatives.

Beyond technical execution, my passion lies in engineering leadership and talent development. As Lab Manager at the UGA Innovation Factory and Director of UGA CREATE Engineering, I executed a critical facility buildout and cultivated specialized technical teams, actively mentoring and managing undergraduate students to scale our advanced internal design engineering capabilities. Furthermore, I am deeply committed to advising the next generation of engineers, having founded the UGA Manufacturing Club and guided its transformation into an officially recognized student chapter of the Society of Manufacturing Engineers (SME).

My background executing custom technical solutions further aligns with the agility required to lead high-growth engineering teams. I also deliver mechatronic prototyping and automation consulting for diverse operational needs, deploying end-to-end solutions that span from ESP32/MQTT sensor topologies to LIDAR scanning and fabrication of iconic historic building facades. I prioritize aggressive technical execution, strategic foresight, and measurable impact.

I am eager to bring this blend of hands-on engineering rigor and proven leadership to your organization to drive both technological innovation and team excellence. I welcome the opportunity to discuss how my experience directly aligns with your immediate technical milestones.

Sincerely,

Marcus DiBattista

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PROFESSIONAL PROFILE

Strategic Engineering Leader and Ph.D. candidate specializing in Cyber-Physical Systems (CPS), Mechatronics, and AI-driven Data Infrastructure. Proven capacity for technical integration, team management, and organizational leadership in high-growth Industry 4.0/5.0 applications. Dedicated to architecting scalable solutions while actively mentoring and developing the next generation of engineering talent.

EDUCATION

Ph.D. Mechanical Engineering | University of Georgia (Expected Aug 2026); GPA: 3.91

- **Dissertation**: Transforming Learning Factories via a Cyber-Physical Co-Design Framework: Methodologies and Applications in Advanced Manufacturing Systems

B.S. Mechanical Engineering | University of Georgia (May 2023); Cum Laude

SKILLS & TECHNICAL INFRASTRUCTURE

- Python, C++, Java, MATLAB, Bash, SQL, Flux, JavaScript, HTML, GCODE
- Linux (NixOS, Ubuntu, Proxmox), ROS, MQTT, Home Assistant, ESPHome, GCP, Arduino, ESP32
- FDM & Resin 3D Printing (Bambu Lab stacks), CNC Machining (Router, Laser, Mill), CAD/CAM (Fusion 360), Welding, Microelectronics, PCB Prototyping, Soldering, UAV Design, Licensed Drone Pilot

EMPLOYMENT & LEADERSHIP

Graduate Research & Teaching Assistant | UGA College of Engineering | May 2023 - Present

- Spearheaded comprehensive infrastructure upgrades on the Innovation Factory Cyber-Physical Assembly Testbed (IF-CAT), drastically boosting IIoT data collection resolution and capabilities
- Guided Capstone engineering teams through complex mechatronics integrations (MiR200/UR-10 Cobots) and process improvements for SMI Composites
- Development Lead for Georgia AI in Manufacturing (Georgia AIM) Project, executing hardware and software development and deployment for federally funded AI initiatives
- Designed and delivered university-level technical lectures on Additive Manufacturing (AM), Lean Production Principles, and Engineering Entrepreneurship
- Authored multiple scholarly articles published in SME Manufacturing Letters

Lab Manager | UGA Innovation Factory | Jan 2022 - Jan 2023

- Managed and mentored undergraduate engineering students in the lab, accelerating their technical growth and expanding advanced internal design engineering capabilities

- Drove project execution and equipment maintenance across robotics, CV, AM, and IIoT domains

Director | UGA CREATE Labs | Jan 2022 - Jan 2024

- Executed a $50K initial lab buildout, aggressively expanding prototyping and manufacturing capacities
- Cultivated and managed a specialized technical team of undergraduate students, rapidly scaling lab bandwidth and domain expertise through hands-on leadership

Founder & Graduate Advisor | SME Chapter S415 | Aug 2025 - Present

- Founded the organization and guided its transformation into an officially recognized student chapter of the Society of Manufacturing Engineers (SME) in September 2026
- Advised and mentored engineering students, facilitating professional development and engagement with advanced manufacturing and leadership principles

Lead Project Management Intern | McKenney's Inc. | June 2020 - August 2021

- Coordinated a highly profitable $500K equipment installation within an active commercial center
- Architected a centralized digital database system to capture and organize maintenance reports

TECHNICAL PROJECTS & RESEARCH

- Developed industrial applications for information-theoretic algorithms for real-time anomaly detection, operator fingerprinting, and 5S compliance quantification in manual assembly
- Engineered an automated computer vision (CV) sorting system optimizing quality control through real-time machine learning inference using YOLO26
- Designed and validated an autonomous, RFID-equipped quadruped leveraging a novel 12 DoF 3D localization algorithm to rapidly monitor inventory in high-density, dynamic environments
- Engineered a standalone additive manufacturing and IoT learning station showing capabilities, opportunities, and integrations for 3D Printing in industry

AWARDS

- SME David Dornfeld Manufacturing Vision Award (2025)
- 1st Place 2025 SME Blue Sky Competition
- AAAS CASE Workshop Award (2025)
- UAS4STEM Drone Search & Rescue (S&R) Competition

PUBLICATIONS

- **ASME Journal of Manufacturing Science and Engineering**
    - Investigation into Visual Entropy for Continuous Monitoring and Anomaly Detection in Manual Assembly Lines (In Progress)
- **SME Journal of Manufacturing Systems**
    - A Tripartite Co-Design Framework for Upgrading Learning Factories into Cyber-Physical Assembly Testbeds (In Progress)
- **SME Manufacturing Letters**
    - Framework for LLM applications in manufacturing
    - SCOUT: An Autonomous UHF RFID-Equipped Robot Dog for Flexible Inventory Monitoring